

Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

Objective 1 Matlab Code and Output:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Throughput (Mbps)

NB_IoT = 0.05*(1-exp(-SNR/8));

LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')

disp([SNR' NB_IoT' LTE_M'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

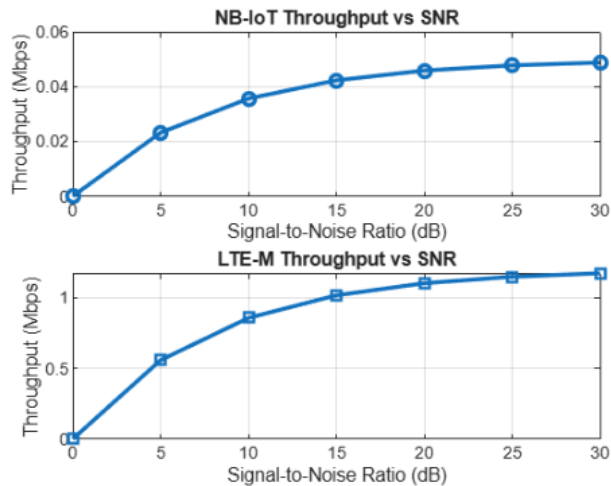
plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')
```

ylabel('Throughput (Mbps)')

15.0000	0.0423	1.0160
20.0000	0.0459	1.1015
25.0000	0.0478	1.1473
30.0000	0.0488	1.1718



grid on

Objective 2 Matlab Code and Output:

```
clc;
```

```
clear;
```

```
close all;
```

```
% SNR values (dB)
```

```
SNR = 0:5:30;
```

```
% Simulated Latency (ms)
```

```
NB_IoT = [180 160 145 130 120 110 100];
```

```
LTE_M = [90 80 70 60 50 45 40];
```

```
disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')
```

```
disp([SNR' NB_IoT' LTE_M'])
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(SNR,NB_IoT,'-o','LineWidth',2)
```

```
title('NB-IoT Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Latency (ms)')
```

```
grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(SNR,LTE_M,'-s','LineWidth',2)
```

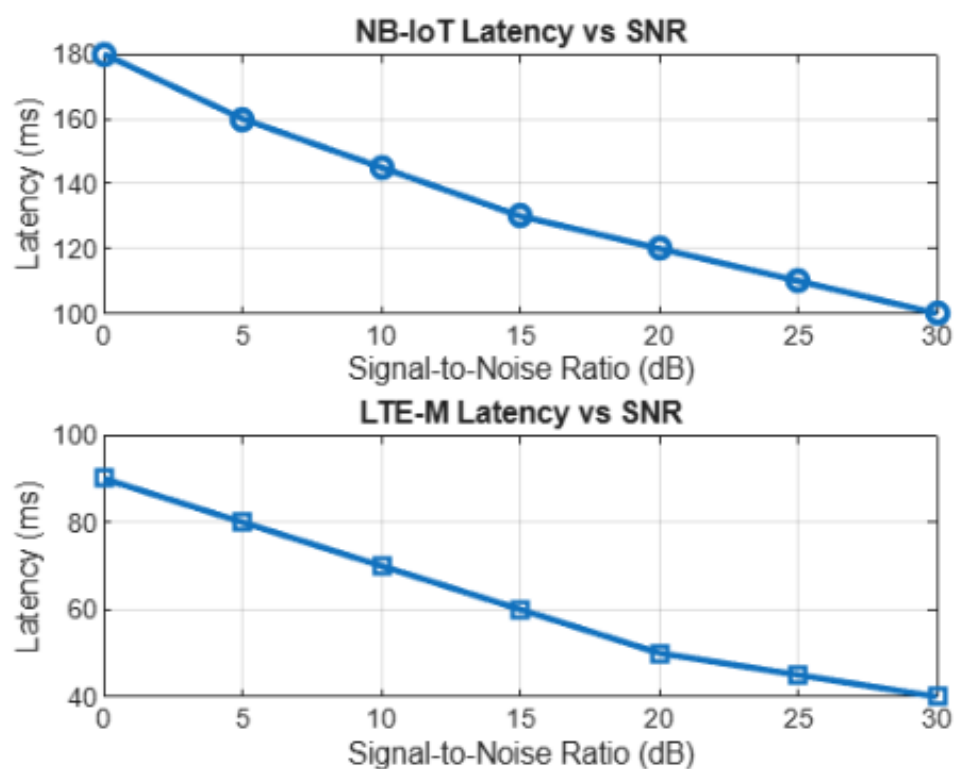
```
title('LTE-M Latency vs SNR')
```

```
xlabel('Signal-to-Noise Ratio (dB)')
```

```
ylabel('Latency (ms)')
```

10	145	70
15	130	60
20	120	50
25	110	45
30	100	40

```
grid on
```



Objective 3 Matlab

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

NB_IoT = [82 86 89 92 95 97 99];

LTE_M = [88 91 94 96 98 99 100];

disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%))')

disp([SNR' NB_IoT' LTE_M'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(SNR,NB_IoT,'-o','LineWidth',2)

title('NB-IoT Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

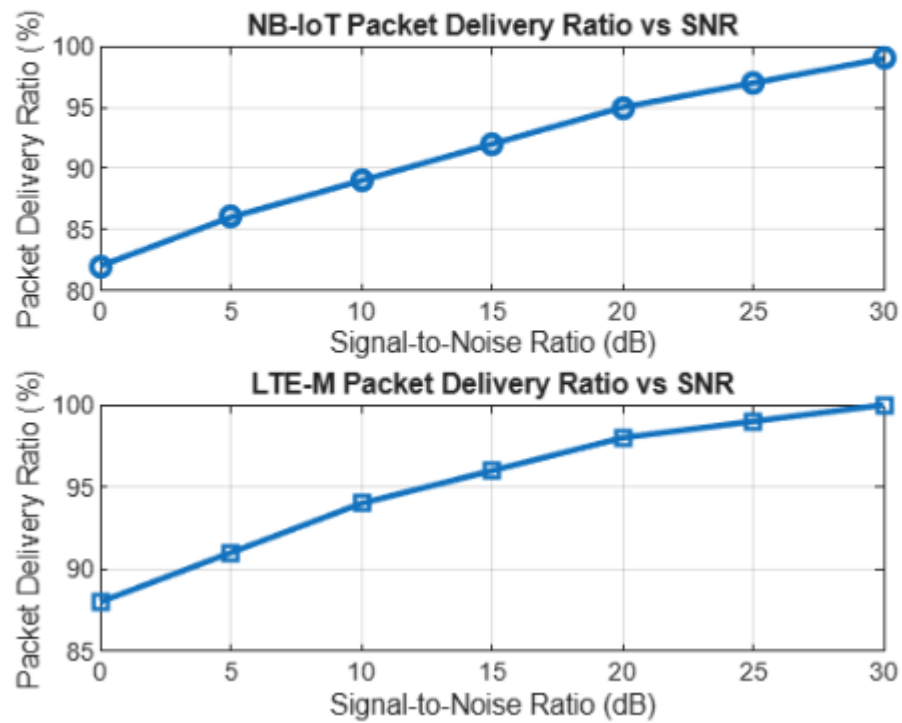
plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Packet Delivery Ratio vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Packet Delivery Ratio (%)')
```

10	89	94
15	92	96
20	95	98
25	97	99
30	99	100



grid on